Cplx

Cplx is a java library leveraging cuda for fast complex polynomial arithmetic over a ring of size N.

It contains four main classes which contain the CPU part of the code and a .cu file containing the kernels. Each class is self-documented but here is a summary:

- **GPUAdd.java** is the basic Add(P1, P2)
- **GPUMatrixAdd.java** is the matrix Add(P, PQ)
- **GPUMultiply.java** is the FFT based basic Mul(P1, P2). It transforms the polynomials in their FFT representation to be able to multiply them in O(n), and then transforms them back to their natural representation.
- **GPUMultiply.java** is the FFT based matrix Mul(P1, P2). It does the same thing as the Multiply version, only on matrices. Each resulting cell is the accumulation of a sum of multiplications, which is computed in-thread in a dedicated kernel. This class is the more advanced since it does slicing (to handle very big matrices) as well as overlapping data copy and execution.

To get a better idea of how things work, what struggles I went through, etc… I recommend reading the **readme.md** and **journal.pdf** at the root of the repository.